

Light-induced photon emission by mammalian cells

Roeland van Wijk^{a,†}, Hans van Aken^a, Weiping Mei^b and Fritz A. Popp^b

^aDepartment of Molecular Cell Biology, State University, Padualaan 8, 3584 CH Utrecht (Netherlands)

^bInstitute of Biophysics, Technology Centre, Opelstrasse 10, 6750 Kaiserslautern 25 (Germany)

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Abstract

In this work, the light-induced photon emission (IPE) by suspensions of mammalian cells was examined. IPE is extremely low and for detection a single photon counting device with a cooled EMI 9558QB photomultiplier tube was used. The mammalian cells in this study were from different tissues and different mammalian species including cat, Chinese hamster, cow, dog, human, monkey, mouse and rat. The IPE was detected in all mammalian cells tested, but was different for the various cell types, ranging from 4 to 100 photons per 10^4 cells. Although our data agree with previous studies in that the IPE of non-fibroblastic normal cells is distinct from that of malignant cells our results reveal that cells of fibroblastic origin show the highest IPE values.

Keywords: Induced photon emission, single photon counting, mammalian cells

1. Introduction

Recent studies have demonstrated re-scattered emission by suspensions of mammalian cells after brief illumination with an ordinary light source [1–4]. This light-induced photon emission (IPE) is extremely weak and special detectors and devices have been constructed for its detection [5, 6]. In the few comparative studies on the IPE of a cultured tumour cell type and its normal counterpart, characteristic differences were observed between the normal and transformed cell, yielding the lowest value for the normal cell and the highest value for the tumour cell [1–4]. However, the quantitative data of these four studies cannot be compared, since the measurement conditions were different in these investigations. We have drawn up an inventory of the mammalian cell types present in our laboratory with respect to their IPE activity. We report the IPE of cells from different tissues and different mammalian species, including cat, Chinese hamster, cow, dog, human, monkey, mouse and rat.

2. Materials and methods

In this study, primary cultures, established normal cell lines and tumour cell lines were used.

[†]Author to whom correspondence should be addressed.

2.1. Primary cultures

Rat hepatocytes were isolated essentially as described previously [7]. For IPE measurements freshly isolated cells were used.

Rat liver fibroblasts were obtained by long-term culturing of the liver cell population after plating at low density. Under these conditions rat liver fibroblasts were able to divide rapidly. After 2 weeks of culture the hepatocytes represented less than 5% of the population, which consisted of 95% fibroblasts. After two further subcultures cells were used for measurement.

Rat heart fibroblasts were obtained from heart cell cultures [8], which were prepared according to a modification of the method of Harary and Farley [9]. Neonatal rats (1–2 days old) were decapitated and the hearts were excised and minced. The mince was incubated in a spinner flask at 37 °C with 0.05%–0.1% trypsin (in 137 mM NaCl, 5 mM KCl, 4 mM NaHCO₃, 5 mM glucose, penicillin (100 000 units l⁻¹) and streptomycin (100 mg l⁻¹). The incubation fluid was decanted and new medium was added. The supernatant from the first three incubations (15 min each) was discarded; during the following 6–8 incubations (10 min each), the mince was almost completely digested. Cell pellets were spun (8 min, 430g) and resuspended in Ham's F10 growth medium (Gibco), supplemented with 10% foetal calf

serum, 10 mM NaHCO₃, penicillin (100 000 units l⁻¹), streptomycin (100 mg l⁻¹), arabinose C (10 µM, to inhibit fibroblast growth) and CaCl₂ (final concentration, 1 mM). The cells were plated on Falcon 3000 dishes for 4 h, during which time fibroblasts adhere and myocytes remain freely suspended. Fibroblasts which remained on the surface of the dish were trypsinized again. They were replated in Dulbecco's Minimal Essential Medium (DMEM, Gibco, supplemented with 10% foetal calf serum) several times and cultured for several weeks before use in the measurements.

Human foreskin fibroblasts were isolated essentially as described previously [10].

2.2. Normal cell lines

The normal cell lines used in this study have been derived from the American Type Culture Collection [11]. The catalogue numbers and the origin of the individual lines are as follows.

(1) CCL1: NCTC clone 929 derived from the parental strain L, which was derived from normal subcutaneous areolar and adipose tissue of a 100-day-old male C3H/An mouse.

(2) CCL34: MDCK (NBL-2) cell line, which was derived from the kidney of a normal, adult, female cocker spaniel.

(3) CCL61: CHO-K1 cells derived as subclone from the parental CHO cell line, which was initiated from a biopsy of an ovary of an adult Chinese hamster.

(4) CCL81: Vero cell line, which was initiated from the kidney of a normal, adult, African green monkey.

(5) CCL92: 3T3-Swiss albino cell line, which was established from disaggregated Swiss mouse embryos; they are contact-sensitive fibroblasts.

(6) CCL226: C3H10T½ clone 8 cells, which were isolated from a line of C3H mouse embryo cells; they are contact-sensitive fibroblasts.

(7) CRL1395: FBHE cell line, which was established from foetal bovine heart endothelial cells.

(8) CRL1658: NIH/3T3 cell line of highly contact-sensitive cells, which was established from NIH Swiss mouse embryo cultures in the same manner as the 3T3 fibroblasts.

2.3. Tumour cell lines

The tumour cell lines used in this study are characterized as follows.

(1) Clone Neuro-2A was established from a spontaneous tumour (neuroblastoma) of a strain A albino mouse (American Type Culture Collection CCL131) [11].

(2) Reuber H35 cell line [12] and HTC cell line [13] were established from transplantable rat hepatomas.

(3) The cat breast carcinoma cell line K248C was obtained from Dr. H. Nederbragt (Department of Pathology, Faculty of Veterinary Sciences, State University, Utrecht, Netherlands).

(4) The human lung tumour cell line SW1573 was originally isolated and characterized as a squamous cell carcinoma by Dr. A. Leibovitz [14].

2.4. Cell culturing

Normal and tumour cells, with the exception of rat hepatocytes, were adapted to growth in DMEM supplemented with 10% foetal calf serum, and buffered with 10 mM *N*-2-hydroxyethylpiperazine-*N'*-2-ethanesulphonic acid (HEPES, pH 7.4). Cell suspensions of monolayers of normal and tumour cells were prepared by trypsinization. After trypsinization the cells were washed and resuspended in DMEM without phenol red for IPE measurements as described previously [3].

2.5. Determination of photon emission

For detection and registration of IPE we used the single photon counting device described previously [3]. This device was equipped with a cooled EMI 9558QB photomultiplier tube. The cathode has a diameter of 44 mm and is sensitive in the range 200–800 nm. The average quantum efficiency in this range was approximately 10%. The phototube output was connected to an amplifier-discriminator and counted by a dual counter. The photomultiplier was cooled to -40 °C which reduced the dark count rate to about 35 counts per second (c.p.s.). It was further equipped with an optical filter unit and a shutter to block the optical path to the sample compartment. A 2 cm quartz cuvette containing 11 ml of cell suspension or medium was placed in the darkened sample compartment at a distance of 125 mm from the photomultiplier. In general, cell densities up to 10⁵ cells ml⁻¹ were used for IPE measurements, with the exception of some cell types which were used at higher cell densities. The sample compartment was kept at 37 °C. Prior to the commencement of the measurements, the samples were kept in complete darkness for 5 min at 37 °C and continuously stirred.

The photon emission intensity was measured and integrated over intervals of 50 ms by computer. For stimulation of photon emission a 150 W halogen tungsten lamp (7158, Philips, Eindhoven, Netherlands) was used. It was situated at right angles to the photomultiplier and was equipped with a

TABLE 1. Induced photon emission of mammalian cells

Species	Cell type	N/T ^a	P/C ^b	IPE per 10 ⁴ cells ^c	OD ₅₅₅ per 10 ⁶ cells ^d
1. Rat	Liver fibroblasts	N	P	114.2 ± 29.8	1.153
2. Cow	Bovine endothelial cells (CRL1395)	N	C	66.5 ± 9.8	0.167
3. Human	Foreskin fibroblasts	N	P	64.4 ± 10.8	0.518
4. Mouse	3T3 fibroblasts (CCL92)	N	C	58.1 ± 5.4	0.174
5. Rat	Heart fibroblasts	N	P	37.2 ± 4.7	0.195
6. Mouse	C3H10T½ fibroblasts (CCL226)	N	C	36.4 ± 9.9	0.430
7. Human	Lung carcinoma SW1573	T	C	36.3 ± 4.8	0.620
8. Mouse	NIH/3T3 (CRL1658)	T	C	28.1 ± 5.1	0.298
9. Rat	Hepatoma HTC	T	C	15.6 ± 1.8	0.287
10. Mouse	Neuroblastoma N ₂ A (CCL131)	T	C	14.9 ± 4.1	0.329
11. Cat	Breast carcinoma K248C	T	C	8.3 ± 1.1	0.144
12. Rat	Hepatoma Reuber H35	T	C	7.6 ± 0.6	0.180
13. Chinese hamster	Ovary cells	N	C	7.6 ± 1.1	0.099
14. Rat	Hepatocytes	N	P	6.1 ± 0.8	0.564
15. Dog	MDCK kidney cells (CCL34)	N	C	5.6 ± 0.4	0.156
16. Mouse	NCTC clone 929 areolar and adipose cells (CCL1)	N	C	5.5 ± 0.8	0.126
17. Monkey	Vero kidney cells (CCL81)	N	C	4.1 ± 1.0	0.098

^aN/T, normal or tumour.^bP/C, primary culture or cell line.^cAverage value (from at least three experiments) and standard deviation.^dOD₅₅₅, optical density at 555 nm.

second filter unit cutting out wavelengths above 720 nm and below 310 nm. The distance between the lamp and the cuvette was 19 cm. No temperature changes were observed during the illumination period. During the pre-illumination the shutter was closed. Each measurement cycle was started by irradiating the sample for 10 s. The shutter was opened after closing the lamp compartment and turning out the lamp; the emission was recorded (and evaluated) from 0.3 to 5.3 s after the end of illumination. Each sample was measured at least three times.

3. Results

Experiments were performed on cell suspensions of cat, Chinese hamster, cow, dog, human, monkey, mouse and rat origin. Several of these cell suspensions were derived from tissues which were disaggregated or kept as primary cultures; others were prepared from cell lines that had been grown in monolayer culture and were considered as established cell lines. The cell lines were either normal, being diploid or slightly hypodiploid, or

tumour. The characteristics are summarized in Table 1.

The decay behaviour of IPE from 0.3 to 5.3 s for the mammalian cell suspensions was similar to that described previously for HTC cells [3]. The decay curves for photon emission decreased continuously and IPE in the period 0.3–1.3 s included 90% of the total IPE, whereas after 5 s the IPE values were similar to the dark count rate obtained without prior illumination. As a measure of IPE intensity we calculated the total amount of photon emission over the measurement period up to 1 s by the accumulation of the 50 ms emission values. In preliminary experiments we detected some IPE after irradiation of an empty cuvette or a cuvette with medium but without cells; selection of quartz cuvettes resulted in a low non-cellular background, which was subtracted in order to obtain the cell-specific IPE values.

By expressing the total number of photon counts in 1 s as a function of cell density, the cells displayed an increasing IPE with increasing cell density (Fig. 1). The IPE per cell was different for the various cell types, ranging from 4 to 100 photons per 10⁴ cells (Table 1).

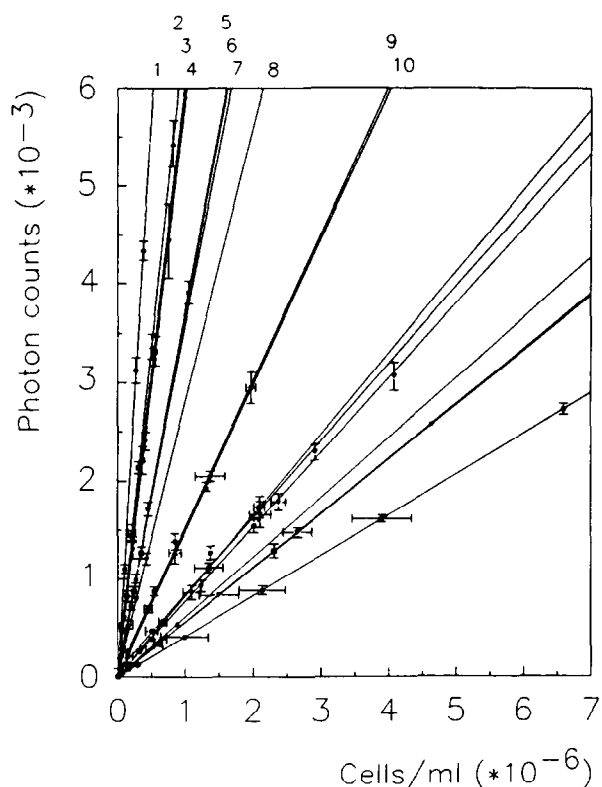


Fig. 1. Cell density dependence of total IPE of suspensions of mammalian cells, after illumination with white light. The total number of photons was calculated by adding the number of photons up to 1 s after illumination. Points represent the average of three determinations of the IPE (error represented by vertical bar) and cell number (error represented by horizontal bar). The numbers outside the figure represent the different cell types and correspond to the numbers given in Table 1.

From these data we believe that IPE is a general phenomenon in suspension of mammalian cells. For comparison, we evaluated the IPE as a function of several cellular characteristics: (1) the species of origin; (2) the type of cell; (3) the degree of transformation. A relationship between the IPE and the species of origin was not observed. Within one species, especially rat and mouse, different cell types show a large variation in IPE. The relationship between IPE and specific cell type is most obvious. Cells of fibroblastic origin, either obtained from primary cultures of dissociated cells or derived from monolayer cultures of established cell lines, show the highest values. The established cell line of foetal bovine endothelial origin also belongs to the group of high IPE activity, suggesting that a high IPE value is characteristic of mesodermal cells. The other cell types tested were either ectodermal or endodermal in origin. The group of cell types of non-mesodermal origin included normal cells and tumour cells. Although in this study tumour cells have been compared with their normal

counterparts in a single case only (liver and hepatoma cells), in general it is obvious that the IPE values of suspensions of normal cells are low, whereas tumour cells have IPE values ranging between the normal and fibroblast-type cells.

Another factor which has been considered to be related to IPE is the size of the cell. From microscopic examination of the cell suspensions it was apparent that the mammalian cells under study differ with respect to their size. In parallel measurements the optical density at 555 nm of the cell suspensions was measured spectrophotometrically at the densities employed in the IPE measurements. The optical density is directly proportional to the cell density under our experimental conditions. The differences in optical density of the suspensions reflect differences in the cell size (Table 1). Cell types were very different with respect to cell size, Vero kidney cells and Chinese hamster ovary cells being the smallest, and hepatocytes and liver fibroblasts being the largest. We found no relation between IPE and cell size. When IPE is expressed relative to optical density, as a measure of cell size, the large differences between the IPE values of different cell types remain.

4. Discussion

The results confirm that suspensions of mammalian cells show a weak light-induced photon emission. The intensity of this emission is dependent on the cell type. In general, the lowest IPE values ranging between 4 and 8 photons per 10^4 cells are found for normal, non-mesodermal, cells. The IPE values for tumour cells range between 7 and 36 photons per 10^4 cells. Relatively high IPE values, between 30 and 100 photons per 10^4 cells, are found in suspensions of fibroblasts. Recently, data on the spontaneous photon emission of normal and tumour tissue have been published [15]. Spontaneous photon emission is lower in normal tissue than in tumour tissue. Since different types of cells, including fibroblasts, are present in isolated tissues, further research is required to compare the light-induced with the spontaneous photon emission of tumour and normal cells.

A consistent explanation for the IPE activity of suspensions of mammalian cells is currently lacking. It was reported that the IPE activity was not retained in the cytoplasmic fraction after cell fractionation. Instead, the IPE activity was found after testing the fraction containing nuclei [3, 4]. However, extracted DNA has no IPE activity [3, 4].

Further understanding of the molecular basis of IPE requires spectral analyses of the activation

of mammalian IPE and of IPE itself. This is difficult due to the low level of photon emission after illumination by white light. Our data allow the selection of other cell types for further analyses. Because this study and others [1–4] show a distinction between mammalian cells, especially normal and tumour cells, this phenomenon deserves further evaluation.

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